

5(a)(i)	any three from: 1 not, made all the time / constitutive; 2 gene switched on / protein made, (only) when needed ; 3 triggered by absence of glucose ; 4 concentration increases when galactose present ;	3
5(a)(ii)	transporter / channel, for galactose or to, allow galactose into cell / increase permeability to galactose ;	1
5(b)(i)	$\frac{2012}{5275} \times 100$; 38(%) ;	2
5(b)(ii)	either any two from mp1, mp2 or mp5 or any two from mp3, mp4 or mp5 <i>either</i> 1 (Msn2 in nucleus) higher when glucose is absent ; 2 (as) Msn2 / transcription factors, bind to DNA in nucleus ; <i>or</i> 3 (Msn2 in nucleus) lower when glucose is present ; 4 (as) Msn2 is not, needed in nucleus / bound to DNA ; <i>plus</i> 5 AVP ; e.g. <i>ref. to</i> diffusion / movement, of tagged Msn2 / unbound Msn2 distributed equally in nucleus and cytoplasm ;	2

Question	Answer	Marks
5(c)	any three from: 1 antibodies are too big to enter cells ; 2 making monoclonal antibodies is, expensive / time consuming ; 3 making monoclonal antibodies harms, mice / animals ; 4 immunofluorescence is only used on dead cells ;	3

Question	Answer	Marks
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